CAPSTONE PROJECT PROPOSAL

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PROJECT: Enterprise Network Design, Implementation and Security
COMPANY: Tartar Enterprise Limited
SOFTWARE: Cisco Packet Tracer

INTRODUCTION

Tartar enterprise is a mid-sized company with Head Office and a Branch Office that require reliable, secure, and scalable network. The company has several departments including Sales, Operations, IT, and Management which needs to communicate while maintaining appropriate network separation and security.
This project proposes the design and implementation of an enterprise network using the Cisco Packet Tracer tool. The network will connect to the Head Office and Branch Office through a WAN link and will use VLANS, routing protocols, access control and more security technologies to provide reliable communications and protect network resources.

PROBLEM STATEMENT

Tartar limited requires a network infrastructure that can:

. Separate departments into logical networks
. Provide reliable communication between the Head Office and Branch Office
. Provide redundancy to reduce network downtime
. Allow room for future expansions
. Prevent unauthorized access and rogue DHCP servers from affecting the network
. Restrict remote management of network devices to authorized IT personnel

PROJECT OBJECTIVES

The main objective for this project is to design, build, and secure a functional enterprise network for Tartar limited. Other objectives include:

1.Design a network connecting both Head Office and Branch Office
2.Create separate VLANS for different departments
3.Develop logical addressing and subnetting scheme

4.Configure inter-VLAN routing and implement DHCP for selected VLANS
5.Configure EIGRP to provide dynamic routing between routers
6.Configure WAN connection between the two offices.
7.Configure ACLs to control unauthorized traffic between departments and implement port security to restrict unauthorized devices
8.Configure DHCP snooping to protect against rogue DHCP servers
9.Configure SSH for secure remote device management and implement HSRP for gateway redundancy
10.Configure EtherChannel for switch-link redundancy
11.Implement Spanning Tree Protocol to prevent layer 2 loops
12.Test and document complete network.

PROPOSED NETWORK DESIGN

Proposed network will consist of:

3 routers – Two at the Head Office and one for Branch Office
3 switches – Two at the Head Office and one at the Branch Office
Several PCs representing users in the different departments
Point-to-Point WAN connection between the Head Office and Branch Office routers

The departments will be separated using VLANs:

VLAN	Department
10	Sales
20	Operations
30	IT
40	Management
99	Network Management

IP ADDRESSSING PLAN

IPV4 addressing scheme will be used and different subnets will be allocated to each department and location. It will use subnetting according to the number of required hosts in each department so that it can minimize address wastage while leaving room for future growth.
The IP Address 10.0.0.0/8 will be used which will be appropriate for an internal enterprise network like Tartar Enterprise. It's a large network which will leave room for future

expansion. The Head office will be assigned 10.10.0.0/16, and the Branch office will be assigned 10.20.0.0/16.
For the WAN addressing, we will use 10.10.254.0/30 which gives us two usable addresses which are suitable for a point- to–point connection. The HSRP address will be the .1 is the virtual HSRP gateway the .2 is HQ-RTR1 redundant gateway and the .3 is theHQ-RTR2 redundant gateway. While switches will be assigned address in VLAN 99, and their default gateways will be the appropriate management for VLAN gateway.

Head Office

The network of the Head office will be10.10.0.0/16 which will be subnetted according to the number of hosts required per department.

VLAN	Department	No of host	IP address	Subnet mask	Default gateway
10	sales	20	10.10.10.0/27	255.255.255.224	10.10.10.1
20	operations	20	10.10.20.0/27	255.255.255.224	10.10.20.1
30	IT	10	10.10.30.0/28	255.255.255.240	10.10.30.1
40	management	8	10.10.40.0/28	255.255.255.240	10.10.40.1
99	Network management	6	10.10.99.0/28	255.255.255.240	10.10.99.1

Branch office

The network of the Branch office will be10.20.0.0/16 which will be subnetted according to the number of hosts required per department.

VLAN	Department	No of host	IP address	Subnet mask	Default gateway
10	Sales	15	10.20.10.0/27	255.255.255.224	10.20.10.1
20	Operations	15	10.20.20.0/27	255.255.255.224	10.20.20.1
30	IT	8	10.20.30.0/28	255.255.255.240	10.20.30.1
40	Management	5	10.20.40.0/29	255.255.255.248	10.20.40.1
99	Network management	4	10.20.99.0/28	255.255.255.240	10.20.99.1

ROUTING DESIGN

EIGRP will be configured as the dynamic routing protocol between routers. Its purpose is to allow routers to automatically learn about the networks available at other locations.

Inter-VLAN routing will also be implemented so that communication between VLANs can occur to the organization's policies.

WAN connection between the Head Office and Branch office will provide communication between the two locations.

NETWORK SECURITY

Several security mechanisms will be implemented to protect the network, this includes:

a. Access Control Lists
Extended ACLs will be used to control traffic between departments. Example Sales users will be prevented from accessing sensitive IT sources.
b. Port Security
Port security will be enabled on selected access ports to restrict which devices can Connect to the network. This helps unauthorized devices from gaining network access.
c. DHCP Snooping
DHCP snooping will be enabled on switches to identify trusted DHCP ports and prevent unauthorized DHCP servers from providing incorrect IP addressing information.
d. Secure Remote Management SSH will be configured for remote management of routers and switches.
e. Device Hardening
Network devices will be configured with encrypted passwords and login banners. Only Authorized IT personnel will be permitted to remotely manage network devices.

NETWORK REDUNDANCY

HSRP will be implemented to provide default-gateway redundancy. If the primary router is Unavailable, the backup router will be able to provide gateway services.

EtherChannel will be configured between switches to combine multiple physical links into logical links to provide link redundancy and improve reliability.

Spanning Tree Protocol (STP) will be used to prevent layer 2 switching loops.

DHCP IMPLEMENTATION

DHCP will be configured for at least one department VLAN to automatically provide clients with:
IP addressing
Subnet mask
Default gateway
DNS information

TESTING AND VERIFICATION

After implementation the network will be tested to ensure that all requirements have been met this will include:
 Ping between devices within same VLAN
 Ping between permitted VLANs
 Connectivity between Head Office and Branch Office
 Verification of EIGRP neighbor relationships
 Verification of routing table
 Verification of VLAN and trunk configurations Testing DHCP address allocation
 Testing ACL restrictions
 Testing port security
 Testing DHCP snooping
 Testing SSH access
 Testing HSRP failover
 Verification of EtherChannel
 Verification of spanning tree

EXPECTED OUTCOMES

At the completion of the project Tartar enterprise is expected to have a functional enterprise network that:

 Provide reliable connectivity between the Head Office and Branch Office
 Separate departments using VLANs
 Provide efficient IP addressing
 Dynamically exchange routes using EIGRP

- Protect against unauthorized network devices
- Prevent rogues DHCP servers
- Provide secure remote management through SSH
- Provide gateway and link redundancy
- Prevent switching loops
- Provide a scalable foundation for future network expansion

CONCLUSION

This project will demonstrate the application of CCNA networking concepts in a realistic enterprise environment proposed network combines VLANs, IP addressing, inter-VLAN routing, EIGRP, WAN connectivity, DCP, ACLs, port security, DHCP snooping, SSH, HSRP, EtherChannel and STP

The final implementation will be created and tested in Cisco Packet Tracer with the completed topology, verifications results and documentation.